Java Training School

Bus Ticket Booking App

Project Requirements & Technical Specification

Project Type	Full-stack bus ticket booking application
Frontend	React, HTML, CSS, JAVASCRIPT
Backend	Java, Spring Boot, REST APIs
Database	PostGreSQL

1. Project Overview

The objective of this project is to design and develop a full-stack bus ticket booking application that enables administrators to manage bus categories and bus records, while allowing customers to register, search buses by route, travel date and time window, compare available options and prices, and book a seat when availability is confirmed.

2. Technology Stack

- Frontend: React
- Backend: Java with Spring Boot
- API Communication: RESTful APIs using JSON
- Database: PostgreSQL
- Testing: JUnit and Mockito for service-layer unit tests

3. Functional Requirements

3.1 Administrator Features

Administrators should be able to perform Create, Read, Update and Delete (CRUD) operations for the following entities:

- Bus Categories: Create, view, update and delete categories such as Non-AC/AC Buses, Luxury Buses and Sleeper Buses, Chair Car Buses.
- Buses: Create, view, update and delete records for different bus brands, along with the category and relevant bus details such as bus no, seating capacity etc.

3.2 Customer Features

- Customer Registration: Allow customers to register and maintain their profile information.

- Starting Station: Allow the customer to select the departure station.
- Destination Station: Allow the customer to select the destination station.
- Travel Date: Allow the customer to select the intended date of travel.
- Time-Window Search: Allow the customer to narrow bus search results by a specified time window.
- Search Results and Prices: Display available bus options matching the search criteria, along with their ticket prices.
- Seat Availability and Booking: Allow the customer to select a bus and book a seat only when seat availability is confirmed.
- If a direct bus is not allowed, show them all the connecting routes.

4. Proposed Database Design

Design a normalized relational schema with suitable primary keys, foreign keys, constraints and indexes. The following tables are recommended as a starting point; additional tables may be added as required.

Some of the example tables are as below:

Table	**Purpose**
Customer	Customer profile and contact details
BusCategory	Bus categories such as Normal, Luxury and Sleeper
Bus / Vehicle	Bus brand, category and bus details
Station	Starting and destination station information
BusSchedule / Trip	Route, departure/arrival time, travel date and scheduled bus
Booking / Ticket	Customer, selected trip, seat, ticket price and booking status

5. Java Code Design

Use object-oriented programming (OOP) principles. Follow a clear package structure with classes such as controller, service, repository, entity, request, response and util wherever necessary. The use of suitable interfaces and abstract classes is highly recommended.

6. Exception Handling and Validation

Implement a consistent exception-handling mechanism, including suitable user-defined exceptions, such as:

- CustomerNotFoundException

- BusNotFoundException
- BusCategoryNotFoundException

7. Unit Testing

Write proper unit test cases for service classes. Use JUnit and Mockito to isolate business logic from database and external dependencies.

~ Requirement Document

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— End of Project Specification —